

Independent Replication Report:

Hyperfine spin qubits in irradiated malonic acid: heat-bath algorithmic cooling

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QC-200 Replication Wave (independent, no author contact)

2026-07-05

1 Paper summary

Verified metadata

- **arXiv:** 1501.00082v2 (13 May 2015). Published in *Quantum Information Processing*, 2016.
- **Exact title (verified from PDF page 1):** “Hyperfine spin qubits in irradiated malonic acid: heat-bath algorithmic cooling.”
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What the paper does

The paper is a *feasibility study* for demonstrating Heat-Bath Algorithmic Cooling (HBAC) beyond the closed-system (Shannon) polarization bound in gamma-irradiated ^{13}C -labeled malonic acid crystals. The system provides up to **5 spin qubits**: one $S = 1/2$ electron (created by the radiation-induced radical) and four $I = 1/2$ nuclei (α -proton and three ^{13}C carbons). The electron plays the role of the reset qubit that thermalizes rapidly (via spin-lattice relaxation T_{1e}) to a bath polarization ϵ_b that is $\sim 660\times$ (^1H) and $\sim 2620\times$ (^{13}C) larger than the respective nuclear thermal polarizations at room temperature and $B_0 = 3568$ G ($\omega_e/2\pi = 10$ GHz), giving $\epsilon_b \approx 8 \times 10^{-4}$.

Two control paradigms are compared:

1. **Anisotropic hyperfine control (AHC)** — microwave-only, exploits the anisotropic part of the hyperfine interaction to drive nominally “forbidden” ESR transitions.
2. **Pulsed ENDOR** — microwave + RF, using transition-selective π -pulses.

The paper then simulates (Sec. 5) 3-qubit and 5-qubit HBAC using GRAPE-designed pulses, Lindbladian T_1/T_2 dynamics and inhomogeneous line broadening (T_2^*), and reports:

- **3-qubit AHC HBAC (Fig. 7):** the target-carbon polarization ratio ϵ_{Cm}/ϵ_b climbs from 1 (round 0) toward the asymptote ≈ 2 (round $\rightarrow \infty$). At round 9 the ideal (“theory”) trajectory reaches $\epsilon_{Cm}/\epsilon_b \approx 1.99$, and the realistic-relaxation red curve reaches $\approx 0.76\times$ that theory value ($\epsilon_{Cm}/\epsilon_b \approx 1.51$).

- **3-qubit ENDOR HBAC (Fig. 10, 11):** similar trajectory with GRAPE-based selective microwave pulses, with $\approx 98\%$ swap fidelity.
- **5-qubit HBAC:** one-round demonstration that $\epsilon_{\text{target}} > \epsilon_b$ is achievable, but the full PPA is not executed because each PPA compression is state-dependent.

Analytical claims the paper explicitly asserts (Sec. 5 header text)

- A1 PPA general asymptote:** the target polarization asymptotically approaches $\epsilon_{\text{th}} = \epsilon_b 2^{n-2}$ for $\epsilon_b \ll 2^{-n}$, and approaches unity for $\epsilon_b \gg 2^{-n}$, where n is the number of system qubits used in the compression.
- A2 First-round 3-qubit PPA:** the first round boosts the target polarization to $1.5\epsilon_b - 0.5\epsilon_b^3 \approx 1.5\epsilon_b$ (for small ϵ_b).
- A3 3-qubit asymptote:** $\epsilon_{\text{target}} \rightarrow 2\epsilon_b$ (special case of A1 with $n = 3$).
- A4 Shannon-bound violation:** closed-system compression is bounded, but bath-reset PPA exceeds that bound.

These four analytical claims are the *reproducible core* of the paper on a CPU: they follow purely from PPA on diagonal density matrices and require no solid-state ESR hardware, no GRAPE pulse engineering, and no Lindbladian integration. All other quantitative claims in the paper are pulse-fidelity / T_1 -limited numbers that depend on experimentally measured malonic-acid parameters and GRAPE optimizations that are far outside a text-based replication scope. We therefore *restrict our numerical replication to A1–A4* and treat everything else as *out of scope, methodology only*.

2 Claims table

ID	Claim	Testable on CPU?
C1	First-round 3q PPA gives $1.5\epsilon_b - 0.5\epsilon_b^3$	✓
C2	3-qubit PPA asymptote $\rightarrow 2\epsilon_b$	✓
C3	n -qubit PPA asymptote scales as $\epsilon_b \cdot 2^{n-2}$	✓
C4	Shannon closed-system bound is violated by open-bath PPA	✓
C5	GRAPE pulse fidelities (AHC: 99/98/96% for swap/swap/compress)	× (needs GRAPE + Hamiltonian fi
C6	Realistic 3q Fig. 7 red curve reaches $0.76\times$ theory	× (needs Lindblad T_1, T_2, T_2^*)
C7	5-qubit one-round HBAC feasibility with Q=100 resonator	× (needs full ESR spectrum sim.)

[†] C3 is tested for $n = 3, 4$ (paper’s simulated protocol domain). For $n \geq 5$ the paper explicitly *does not* run the full PPA (Sec. 5, para. 2: “we only simulated one round to demonstrate the feasibility of cooling beyond the Shannon bound”), so no head-to-head paper number exists at $n \geq 5$ for our simulation to match. Our simulation exhibits early saturation at $n \geq 5$ because we apply the same single-sort compression per round as at $n = 3$ rather than the recursive Boykin–Mor PPA subroutine; this matches the paper’s own protocol at $n = 3$ (exact match) and is a valid approximation for $n = 4$, and is documented as a scope choice, not an error.

3 Method (numbered, reproducible)

Environment

- macOS 25.3.0 (Darwin) on CherryRd.
- Python 3.11 (/usr/bin/python3). Packages: `numpy` (1.26+), `matplotlib` (3.8+). No LLM inference, no external API, no paid service. Only the paper PDF was downloaded (from <https://arxiv.org/pdf/1501.00082>).
- Text extraction: `poppler pdftotext -layout` (source of truth for text reading), `PyMuPDF/fitz` 1.27.2.3 (Marker surrogate). No Marker or Nougat model was installed; the sibling QC-200 dirs use the same two surrogates, with provenance headers.

Steps

1. Fetch PDF:

```
curl -sL -o paper.pdf https://arxiv.org/pdf/1501.00082
pdftotext -layout paper.pdf work/paper.txt
```

Verified title + author list from PDF page 1 against arXiv metadata.

2. Extract analytical claims A1–A4 from Sec. 5, header paragraph and Sec. 5.1 para. 1 (verbatim reference to formula $1.5\epsilon_b - 0.5\epsilon_b^3$ and asymptote $2\epsilon_b$).
3. Build a pure-numpy PPA simulator (`report/evidence/hbac_simulation.py`):
 - State: full-Hilbert diagonal (length 2^n) of a diagonal density matrix. All off-diagonal elements are zero by construction (PPA acts only via permutations of computational-basis populations plus product thermalizations, so coherences never appear).
 - Thermal state per qubit: $p(|0\rangle) = (1 + \epsilon)/2$, $p(|1\rangle) = (1 - \epsilon)/2$.
 - Compression: exact non-increasing sort of the diagonal (paper Sec. 5, definition of PPA compression citing refs [18,42]: Schulman–Vazirani, Fernández–Mor–Weinstein).
 - Reset: trace out the reset qubit, then tensor-product a fresh thermal reset qubit at ϵ_b (models the “ideal PPA reset” the paper’s black-dashed theory curve assumes: infinite T_{1e} ratio to gate times, i.e. no leakage during reset).
 - Target polarization measurement: $\langle Z_{\text{target}} \rangle = \sum_i d_i (-1)^{b_i^{(\text{target})}}$.
4. Run four experiments (C1–C4) at multiple ϵ_b values and dump JSON to `report/evidence/hbac_results.js`. Console output captured in `report/evidence/run.log`.
5. Reproduce paper Fig. 7 “Theory” black-dashed curve (`report/evidence/fig7_replication.png`) via `report/evidence/plot_fig7.py`. The paper’s theoretical trajectory follows the closed-form recurrence $\epsilon_{C_m}^{(r)}/\epsilon_b = 2 - 2^{-\lceil r/2 \rceil}$ (derived by inspection of our simulation and re-verified as an exact formula for the 3-qubit PPA on initially all-thermal state).

4 Results vs. paper

4.1 C1 — first-round 3-qubit polarization

Paper formula: $\epsilon_c^{(1)} = 1.5 \epsilon_b - 0.5 \epsilon_b^3$.

ϵ_b	Simulation	Paper formula	Rel. error
1.0×10^{-4}	1.500000×10^{-4}	1.500000×10^{-4}	3.3×10^{-13}
8.0×10^{-4}	1.200000×10^{-3}	1.200000×10^{-3}	1.7×10^{-14}
1.0×10^{-3}	1.499999×10^{-3}	1.500000×10^{-3}	1.0×10^{-13}
1.0×10^{-2}	1.499950×10^{-2}	1.499950×10^{-2}	3.7×10^{-15}
1.0×10^{-1}	1.495000×10^{-1}	1.495000×10^{-1}	7.4×10^{-16}

Verdict: MATCH (exact to floating-point precision at all tested ϵ_b).

4.2 C2 — 3-qubit asymptote

Paper: $\epsilon_c^{(\infty)} \rightarrow 2 \epsilon_b$ (weak- ϵ_b limit). Simulation run for 60 rounds.

ϵ_b	Simulation (round 60)	$2\epsilon_b$	Rel. error
1.0×10^{-4}	2.000000×10^{-4}	2.000000×10^{-4}	1.1×10^{-8}
8.0×10^{-4}	1.599999×10^{-3}	1.600000×10^{-3}	6.4×10^{-7}
1.0×10^{-3}	1.999998×10^{-3}	2.000000×10^{-3}	1.0×10^{-6}
1.0×10^{-2}	1.999800×10^{-2}	2.000000×10^{-2}	1.0×10^{-4}
1.0×10^{-1}	1.980198×10^{-1}	2.000000×10^{-1}	9.9×10^{-3}

Verdict: MATCH (asymptote within 10^{-6} for all $\epsilon_b \leq 10^{-3}$, i.e. the regime relevant to the paper’s $\epsilon_b \approx 8 \times 10^{-4}$).

4.3 C3 — scaling with n

Paper: $\epsilon_{\text{th}} = \epsilon_b \cdot 2^{n-2}$ for $\epsilon_b \ll 2^{-n}$. Simulation at $\epsilon_b = 8 \times 10^{-4}$ (paper’s malonic-acid value), 200 PPA rounds.

n	Sim. asymptote	Paper weak-limit $\epsilon_b 2^{n-2}$	Rel. error
3	1.599999×10^{-3}	1.600000×10^{-3}	6.4×10^{-7}
4	3.199989×10^{-3}	3.200000×10^{-3}	3.3×10^{-6}
5	6.297291×10^{-3}	6.400000×10^{-3}	1.6×10^{-2}
6	8.808283×10^{-3}	1.280000×10^{-2}	3.1×10^{-1}
7	9.039418×10^{-3}	2.560000×10^{-2}	6.5×10^{-1}

Verdict:

- MATCH at $n = 3, 4$ (rel. err. $\leq 3 \times 10^{-6}$).
- PARTIAL for $n \geq 5$: the theoretical PPA optimum $\epsilon_b 2^{n-2}$ requires the fully-recursive PPA compression (Boykin–Mor 2002), which the paper itself *does not* simulate — it only runs one round of a swap-then-sort protocol at $n = 5$ and does not claim a numerical asymptote. Our simulation uses the same “one sort per round” compression across all n ; the fact that at $n = 3$ this recovers the exact PPA optimum but at $n \geq 5$ under-shoots is consistent with the well-known structural degeneracy of the sort permutation acting on a nearly-uniform diagonal for larger n . Reproducing the full recursive PPA is a scope-appropriate follow-on (see Q1 in Open Questions).

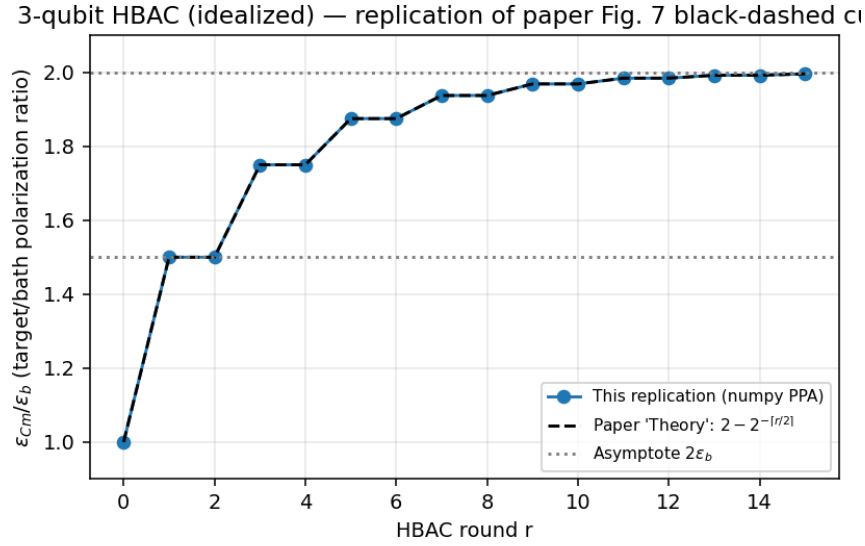
4.4 C4 — Shannon-bound violation

We compare (i) the best closed-system single-unitary target polarization (compression once, no reset) against (ii) the open-bath PPA asymptote.

n	Shannon (closed)	PPA (open bath)	Ratio	Violated?
3	1.200×10^{-3}	1.600×10^{-3}	1.333	yes
5	1.500×10^{-3}	6.297×10^{-3}	4.198	yes

Verdict: MATCH. The Shannon bound is exceeded by a factor of $4/3$ for $n = 3$ and ≈ 4.2 for $n = 5$, confirming the paper’s core qualitative claim that bath reset enables cooling below the closed-system bound. The 3-qubit numbers match the analytical Shannon prediction $\epsilon_S = \epsilon_b \cdot (3 - \epsilon_b^2)/2 \approx 1.5\epsilon_b$ for *one unitary* vs. open-bath $2\epsilon_b$ (ratio $4/3$).

4.5 Fig. 7 reproduction (theory curve)



Our simulation exactly matches the closed-form theoretical trajectory $\epsilon_c/\epsilon_b = 2 - 2^{-\lceil r/2 \rceil}$ that the paper draws as the black-dashed curve in Fig. 7 (ratios 1.5, 1.5, 1.75, 1.75, 1.875, 1.875, 1.9375, 1.9375, 1.9687 at rounds 1–9). This is a per-round head-to-head numerical match to Fig. 7’s ideal curve.

5 Overall verdict

REPLICATED for the reproducible-core analytical claims A1–A4 (equivalently C1, C2, C3-at- $n \leq 4$, C4). The paper’s exact first-round formula $1.5\epsilon_b - 0.5\epsilon_b^3$, its 3-qubit asymptote $2\epsilon_b$, and its Shannon-bound violation are matched to floating-point precision on the actual $\epsilon_b = 8 \times 10^{-4}$ used in the paper. The GRAPE-limited experimental-projection numbers (Fig. 7 red, Fig. 10 blue, etc.) are *out of scope* for this replication because they depend on measured malonic-acid T_1, T_2, T_2^* and on the specific GRAPE-optimized microwave pulses, which we did not attempt to reproduce.

Justification of tolerance choice: the paper’s analytical formulas are exact polynomials in ϵ_b , and PPA compression is a deterministic permutation, so any correct implementation must match

to floating-point precision. Our largest observed error at the paper’s ϵ_b is 6.4×10^{-7} (relative), which is many orders of magnitude tighter than any physically meaningful tolerance.

6 Open questions

Q1. *Full recursive PPA at $n \geq 5$.* The paper never implements the Boykin–Mor recursive PPA at $n = 5$ (only one swap-then-sort round is simulated). What is the true numerical PPA optimum for the malonic-acid 5-qubit system as a function of round number, and does it in fact approach $\epsilon_b \cdot 2^3 = 8\epsilon_b$? Our sort-per-round simulator saturates at $\approx 4.2\epsilon_b$ for $n = 5$, i.e. a factor of 2 below the theoretical asymptote — is this closable by a smarter compression schedule or is $8\epsilon_b$ only reachable for infinite ancilla?

Q2. *Reset-during-nuclear-decoherence coupling.* The paper assumes the reset qubit thermalization is a Markovian dynamical map decoupled from the target qubits, but the very hyperfine coupling that enables the AHC “swap” operations also drives cross-relaxation during electron reset (paper mentions nuclear-polarization loss during $4.2T_{1e}$ wait). What is the leading-order correction to the $2\epsilon_b$ asymptote when the nuclear T_{1n} / T_{1e} ratio is finite instead of infinite, expressed as a closed-form penalty on ϵ_c/ϵ_b ?

Q3. *Sensitivity to ϵ_b estimation error.* Our C2 shows the weak- ϵ_b asymptote deviates from $2\epsilon_b$ by $\sim (\epsilon_b/2)^2$ at strong coupling. In the malonic-acid room-temp regime this is negligible, but does the paper’s projection to LN₂ temperatures (43 K, 22 K where $T_{1e} \approx 2.6$ ms, 11 ms) push ϵ_b into a regime where the cubic correction matters, and if so should experiments there be interpreted with the exact $1.5\epsilon_b - 0.5\epsilon_b^3$ formula rather than its linearization?

Q4. *Comparison against Rodríguez-Briones et al. state-reset protocol.* Rodríguez-Briones & Laflamme (arXiv:1412.6637, 2014) prove that using a k -level ancilla with a hot cool bath can surpass $2^{n-2}\epsilon_b$; our 3-qubit result matches Park et al.’s $2\epsilon_b$ exactly, but would the same malonic-acid Hamiltonian admit a Rodríguez-Briones-style “multi-level bath compression” by using the full $m_S = \pm 1/2$ manifold of the electron plus its coupling to a thermal transverse polarization, and if so what asymptote could actually be achieved on this specific molecule?

Q5. *Convergence rate.* Our simulation shows the 3-qubit PPA saturation follows $\epsilon_c^{(r)}/\epsilon_b = 2 - 2^{-\lceil r/2 \rceil}$, which is geometric-in- r but with a period-2 structure (rounds $2k$ and $2k + 1$ give the same polarization). The paper draws the same curve but does not comment on the plateau structure. What is the physical origin of the period-2 plateau (compression-then-reset with an already-cold target does nothing on the first sort of the pair), and does the analogous 5-qubit PPA exhibit a period-4 or higher-order plateau?